

WORK IN PROGRESS

Excelakuten

Automated repair for broken Excel files.

Upload — get back a repaired, validated file.

- 7-stage pipeline: load → diagnose → structure → clean → formula repair → validate → export
- Built-in changelog: every touched cell, before/after + reason
- Fixes what it can — honestly reports what it can't

Python · FastAPI · openpyxl · LibreOffice · Docker-ready

THE PROBLEM

Excel files nobody dares to touch

- #REF! errors and formulas pointing at deleted sheets and dead network drives
- Circular references that make Excel refuse to calculate
- Hidden sheets and columns that visible formulas silently pull numbers from
- Hardcoded values pasted straight over live formulas
- Text dates, numbers stored as text, stray whitespace — looks like data, isn't
- Corrupt XML: the file won't even open

My test file: 106k+ formulas, 28k volatile, three sheets Excel itself wants to "repair".

Seven stages — each one logged

1

Load

parses even corrupt files — CSV, .xls, .ods

2

Diagnose

error map: hidden sheets, external links, broken chains

3

Structure

finds tables, data types, data sources

4

Clean

text dates → dates, "1 234.56" → numbers, dupes out

5

Repair

#REF!, circular refs, pattern-based fill

6

Validate

LibreOffice recalcs everything — new errors = stop

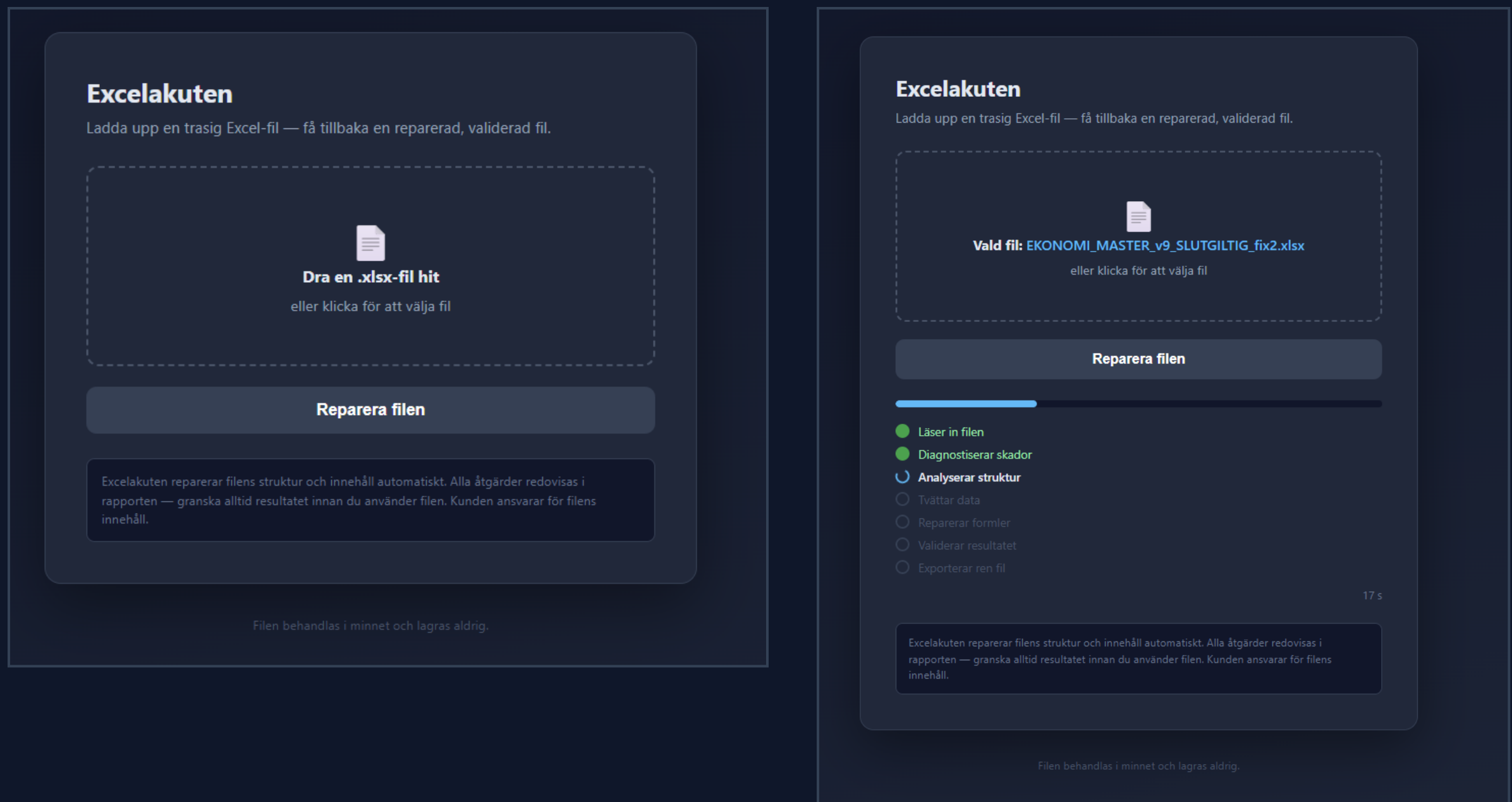
7

Export

clean .xlsx + changelog tab listing every touched cell

THE DEMO

Drag in a file — watch the stages live

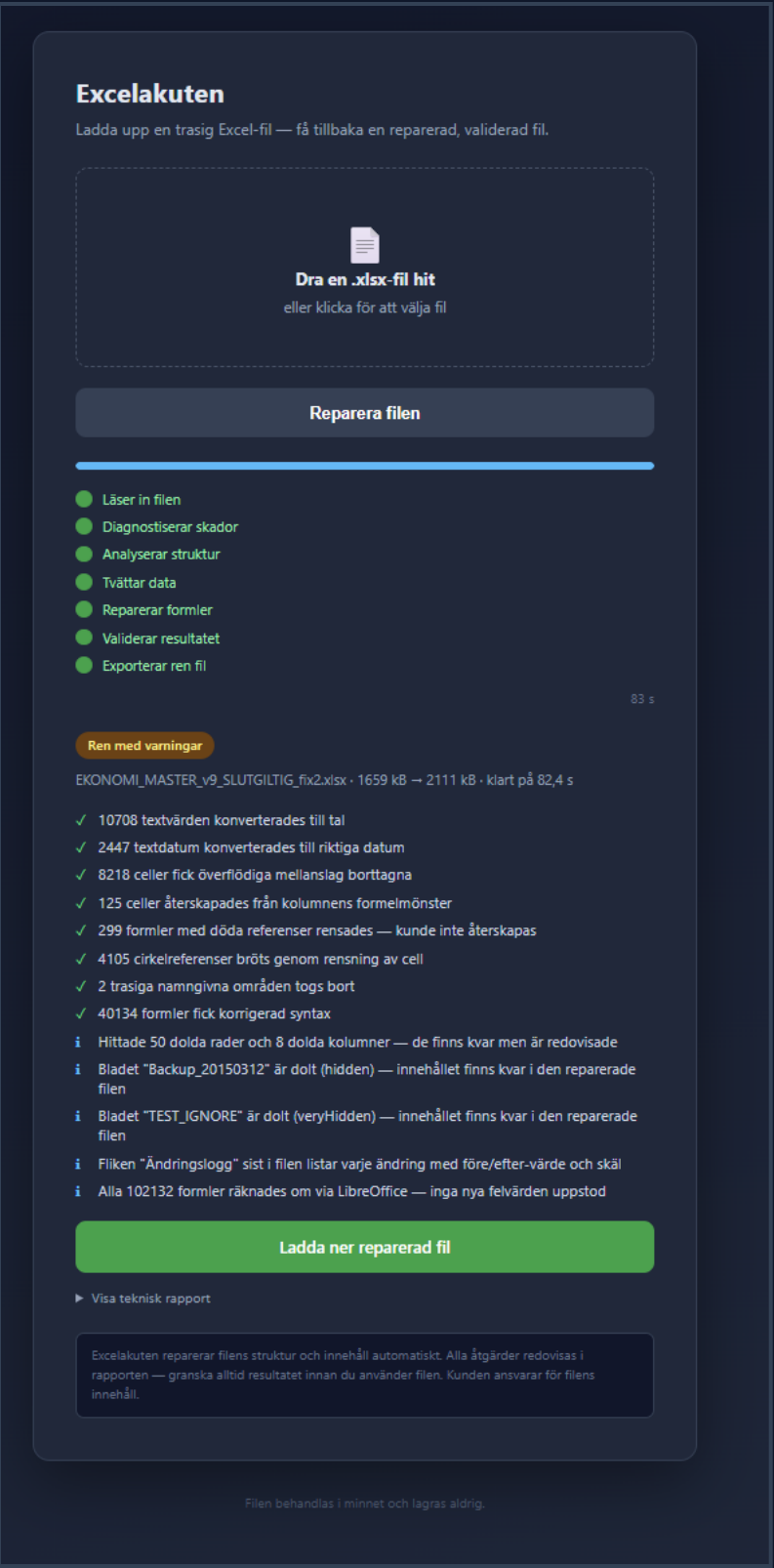


Every stage reports in real time via the job API — no magic spinner.

Files are processed in memory and never stored.

THE RESULT

82 seconds — everything accounted for



10,708

text values → numbers

2,447

text dates → real dates

4,105

circular references broken

40,134

formulas syntax-fixed

102,132

formulas recalculated — 0 new errors

WHY I BUILT IT

It tells you what it CAN'T fix

83 s

Ren med varningar

Kvarstående problem

X

Kolumner med blandade datatyper kvar (col_B, col_B, col_B, col_B)

X

Formler hämtar data från externa filer (Budget_2013.xls) (651 st)

X

Dolda blad i arbetsboken (2 st)

X

Celler där formler ersatts med fasta värden (2458 st)

Remaining problems get flagged

ÄNDRINGSLOGG — Excel Repair Pipeline						
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Varje ändrad cell listas med före/efter-värde och skäl. Typ 'gissning' = återskapat/ersatt värde som bör granskas av en människa.						
Blad	Cell	Före	Efter	Ändrad	Typ	
START LÄS MIG	A15	extern konsult (hette nåt med J7), praktikanter	extern konsult (hette nåt med J7), praktikanter	överflödiga mellanslag borttagna	säker	
START LÄS MIG	A16	chefen (han ändrade ALLT en fredag, återställd	chefen (han ändrade ALLT en fredag, återställd	överflödiga mellanslag borttagna	säker	
Data	K2	=IF(F2="OK",D2,0)	=IF(F2="OK",D2,0)	argumentseparator normaliserad (->.)	säker	
Data	L2	=VLOOKUP(A2:\$A\$1:\$T\$14000;3;FALSE)	=VLOOKUP(A2:\$A\$1:\$T\$14000;3;FALSE)	formel rensad	säker	
Data	N2	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	argumentseparator normaliserad (->.)	säker	
Data	O2			cell rensad	säker	
Data	Q2	=IFERROR(D2/B2,"fel")	=IFERROR(D2/B2,"fel")	argumentseparator normaliserad (->.)	säker	
Data	B3	2016.04.09	2016-09-04T00:00:00	text konverterad till datum	säker	
Data	N3	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	argumentseparator normaliserad (->.)	säker	
Data	Q3	=IFERROR(D3/B3,"fel")	=IFERROR(D3/B3,"fel")	argumentseparator normaliserad (->.)	säker	
Data	L4	=VLOOKUP(A4:\$A\$1:\$T\$14000;3;FALSE)	=VLOOKUP(A4:\$A\$1:\$T\$14000;3;FALSE)	formel rensad	säker	
Data	N4	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	argumentseparator normaliserad (->.)	säker	
Data	Q4	=IFERROR(D4/B4,"fel")	=IFERROR(D4/B4,"fel")	argumentseparator normaliserad (->.)	säker	
Data	B5	24-12-2014	2014-12-24T00:00:00	text konverterad till datum	säker	
Data	D5	88 212.50	88212.5	text konverterad till tal	säker	
Data	I5	4310	4310	text konverterad till tal	säker	
Data	L5	=VLOOKUP(A5:\$A\$1:\$T\$14000;3;FALSE)	=VLOOKUP(A5:\$A\$1:\$T\$14000;3;FALSE)	formel rensad	säker	
Data	N5	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	argumentseparator normaliserad (->.)	säker	
Data	Q5	=IFERROR(D5/B5,"fel")	=IFERROR(D5/B5,"fel")	argumentseparator normaliserad (->.)	säker	
Data	N6	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	argumentseparator normaliserad (->.)	säker	
Data	O6			cell rensad	säker	
Data	Q6	=IFERROR(D6/B6,"fel")	=IFERROR(D6/B6,"fel")	argumentseparator normaliserad (->.)	säker	
Data	I7	4310	4310	text konverterad till tal	säker	
Data	N7	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	argumentseparator normaliserad (->.)	säker	
Data	O7			cell rensad	säker	
Data	Q7	=IFERROR(D7/B7,"fel")	=IFERROR(D7/B7,"fel")	argumentseparator normaliserad (->.)	säker	
Data	I8	4310	4310	text konverterad till tal	säker	
Data	K8	=IF(F8="OK",D8,0)	=IF(F8="OK",D8,0)	argumentseparator normaliserad (->.)	säker	
Data	N8	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	argumentseparator normaliserad (->.)	säker	
Data	Q8	=IFERROR(D8/B8,"fel")	=IFERROR(D8/B8,"fel")	argumentseparator normaliserad (->.)	säker	
Data	N9	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	argumentseparator normaliserad (->.)	säker	
Data	Q9	=IFERROR(D9/B9,"fel")	=IFERROR(D9/B9,"fel")	argumentseparator normaliserad (->.)	säker	
Data	I10	4310	4310	text konverterad till tal	säker	
Data	N10	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	=OFFSET(\$D\$1:ROW()-1,0)+RAND()*0	argumentseparator normaliserad (->.)	säker	
Data	Q10	=IFERROR(D10/B10,"fel")	=IFERROR(D10/B10,"fel")	argumentseparator normaliserad (->.)	säker	
Data	B11	2017.11.09	2017-09-11T00:00:00	text konverterad till datum	säker	

The changelog: every cell, before/after

- Nothing is deleted silently — cleared cells are logged by reference
- The "Ändringslogg" tab in the output: before/after + reason, per cell
- An integrity check diffs every cell against the original — unexplained changes get flagged

Built as Tier-1 of a larger vision:

understand the file's data sources —
then automate the whole pipeline behind it.